CSV Datasource Management

Brainwave GRC

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Introduction

Context

This document presents the CSV datasource management add-on.

This documentation provides information regarding both the initial setup of the add-on as well as its usage by the business users. For a more general introduction to Brainwave and the configuration of the solution, please refer to the full documentation, available using the following URL: https://documentation.brainwavegrc.com.

Executive summary

Once Brainwave has been deployed, it correlates tons of different datasources to provide a 360° view of the company situation:

- Who is working for the company
- Who can access to what
  - at an infrastructure level
  - at an application level
  - at a data level
- Where the problems are located

Technically speaking, because of the company constraints, it often ends up with an hybrid integration situation:

- Some data sources are directly managed by Brainwave through connectors or extraction scripts
  - Active Directory
  - Cloud based apps such as ServiceNow
  - SAP
  - …
- Some data sources are automatically extracted by the customer himself in the form of flat files and pushed to a data dock (such as a shared folder)
- Some data sources are provided directly by the application owners on a regular basis through different means (sharefolder, email, …)

At the end of the day, from an audit and compliance perspective, the resource owners remain responsible for providing relevant data for their applications. We can easily understand here that if it is the case `by design` for the two first use cases (connectors & automated extraction) this is not the case for the later.
It often burden the Brainwave application owner with unnecessary gentle reminders and data quality checks. Furthermore, as this is a manual process, we cannot ensure the auditors that those files have not been altered.

The goal of this optional IAS add-on is to provide an off-the-shelf solution for this use case.
Once installed and properly configured, the application owners will be able to provide their application data extraction directly through the Brainwave web interfaces.

You will also be able to enforce the application data extraction formats at upload time, therefore, ensuring that data quality goals are respected.
Once uploaded, this add-on will maintain a full chain of custody for your auditors: who has provided what and when as well as a SHA256 footprint of the data.
Finally, you will be able to monitor the current upload progress across your application owners and automatically send gentle reminders by email when necessary.

> **Important:** This add-on is used only with applications that provide their information in the form of a single CSV file. It cannot be used with other file formats, nor with applications who provide information as a series of files.
> **Important:** Applications already collected in the project **cannot** be configured via this add-on. It is used only for new applications (not already in the ledger)

Usage

In practice, the iGRC administrator needs to go through a few steps before the upload can be handled by application owners for a new CSV application:

1. As an administrator, in the web portal:
    - Declare a new CSV application
    - Configure the file format
    - Upload a first CSV file for that application
2. As an administrator, using the batch:
    - Launch a new timeslot that includes the add-ons silos
    - This will create the a new application in the Ledger
3. As an administrator, in the portal:
    - Set an owner for the new CSV application (technical or business owner)

> The application owner can now upload new CSV files to the portal for this application.

Setup

Once the add-on is installed on your project, it is available through the `bw_datasourcemanagement` subdirectory.
A series of configuration variable are provided, you **MUST** fullfil them in your technical configuration, otherwise this add-on won't work properly.

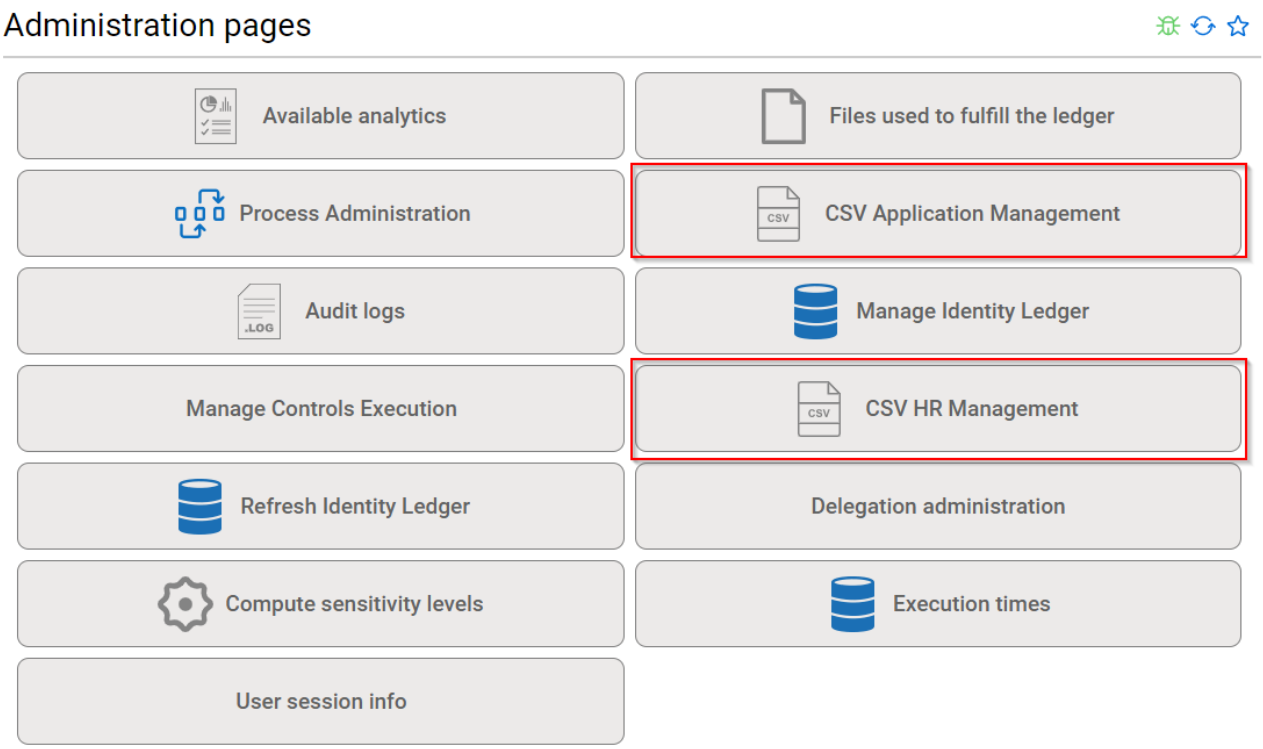
Variable name	Variable description	Typical value
`datasourcemanagement_location`	datasource management base location	{config.projectPath}/importfiles/datasourcemanagement/source/

Variable name	Variable description	Typical value
`datasourcemanagement_pivotlocation`	datasource management pivot files location	{config.projectPath}/importfiles/datasourcemanagement/pivot/
`datasourcemanagement_sendersurname`	Sender user name when sending gentle reminder by email	Security Team
`datasourcemanagement_senderemail`	Sender email when sending gentle reminder by email	security@acme.com
`datasourcemanagement_ccmail`	mail in cc when sending gentle reminder by email	compliance_dataroom@acme.com
`datasourcemanagement_loadpath`	iGRC path to load data	C:/Program Files/igrcanalytics/
`datasourcemanagement_loadbatch`	batch to load data	igrc_batch.cmd
`datasourcemanagement_loadprojectname`	project name to load data	
`datasourcemanagement_loadtechnical`	technical configuration to use to load data	
`datasourcemanagement_loadconfpath`	configuration path to use to load data	C:/Program Files/igrcanalytics/

> **Warning:** `datasourcemanagement_location` and `datasourcemanagement_pivotlocation` must point to a **VALID** location, the add-on won't create any subdirectories on your behalf.

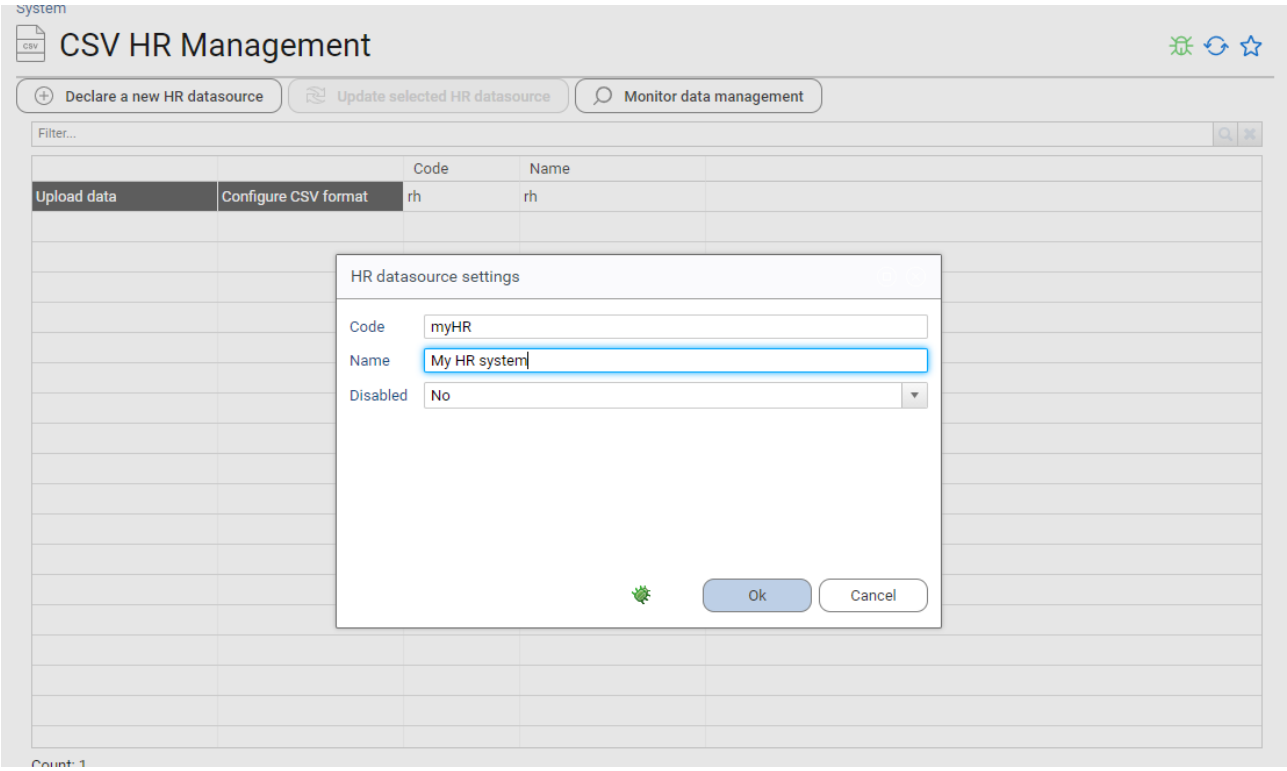
Configuration

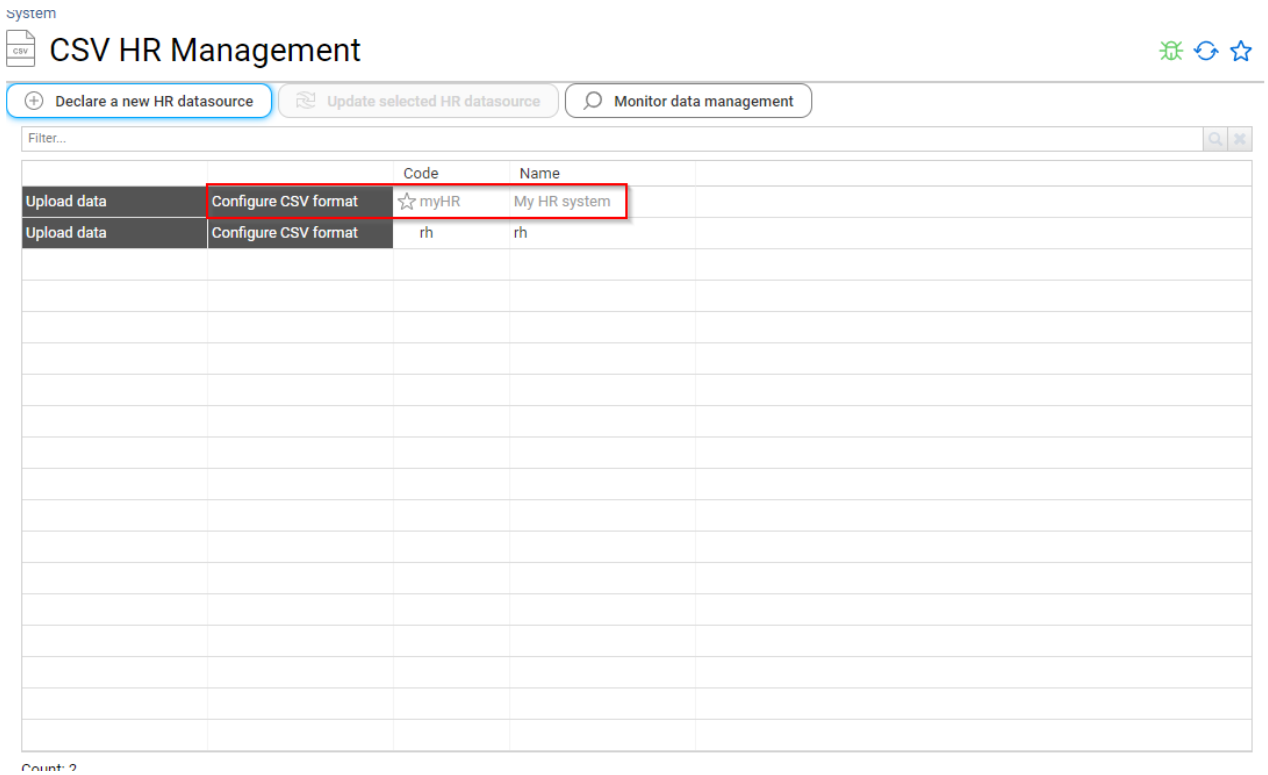
Once properly installed, all configuration is done through the web portal.
You must connect as a `technical administrator` to configure data upload for your data.
You can both configure HR datasources and application datasources. In order to configure your datasources, you first have to go to `System`/`Settings` and select either `CSV HR Management` or `CSV Application Management`:



Configure a new datasource

Go to `CSV HR Management` or `CSV Application Management` and select `Declare a` **`new`** `HR datasource`:



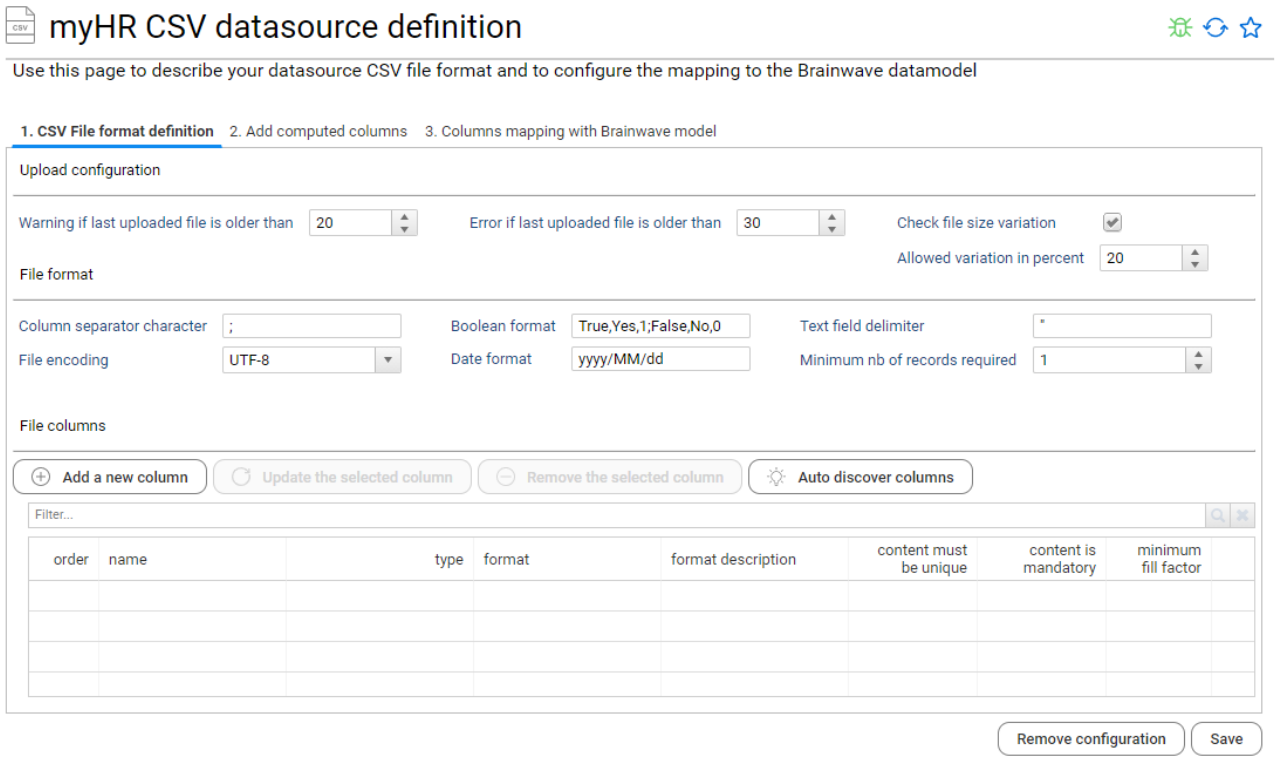


This new datasource appears in gray with a small star:

- gray = This datasource has not been configured yet, it is considered disabled
- star = This is a new datasource, it is not loaded in the Identity Ledger yet

Click on `Configure CSV format` and `Enable file upload`.

You can now configure your datasource:



Configuring a new datasource is a 2 step process:

- You declare your datasource CSV format

- You map your datasource CSV columns to the Brainwave datamodel

An optional step can consist of dynamically computing columns to extract relevant information from the CSV datasource (more on this on `Add computed columns`).

CSV file definition

Though this section you will configure the CSV file format:

- **Upload configuration** - What are the expectations in terms of data freshness and data quality checks at a file level
- **File format** - What is the overall CSV file format
- **File columns** - What are the expected columns

Upload configuration

Section	Configuration item	Description
Upload configuration	Warning if last uploaded file is older than	Is used by the monitoring functions to trigger a gentle reminder by email when the latest validated and uploaded file is older than (in nb days)
Upload configuration	Error if last uploaded file is older than	Is used by the monitoring functions to trigger a gentle reminder by email when the latest validated and uploaded file is older than (in nb days)
Upload configuration	Check file size variation	Enable file check variation, when enabled candidate file size is checked against the latest validated uploaded file. If the deviation is greater than the configured value, the file is rejected with an error. It is useful to enable this check once a series of files have been provided and you consider that the file size does not move a lot between updates. Therefore, detecting truncated/corrupted files
Upload configuration	Allowed variation in percent	This parameter is used in conjunction with the previous one to configure the allowed deviation as a percent of the latest validated uploaded file. For instance if you configure 30% and the latest uploaded file is 100kb, the candidate file will pass the test if it is between 70kb and 130kb
Upload configuration	Allowed variation in percent	This parameter is used in conjunction with the previous one to configure the allowed deviation as a percent of the latest validated uploaded file. For instance if you configure 30% and the latest uploaded file is 100kb, the candidate file will pass the test if it is between 70kb and 130kb

File format

Section	Configuration item	Description
File format	Column separator character	The character used as a column separator, note that special characters such as Tab can be configured as well here with the following notation \t
File format	File encoding	The file encoding. This parameter is a bit tricky as the engine will try to detect the file encoding through the data provided, it can results with false positive, to avoid checks just select `ascii`
File format	Boolean format	Used to detect if a **`boolean`** `column` contains allowed values. You start by declaring all allowed values for True separated by comma without any spaces, you add a semi-column and declare all allowed values for False as a comma separated string without any spaces. An example can be `True,1,Yes;False,0,No`
File format	Date format	Used to detect if a `date column` contains allowed values. This field contains a date description with a special string format described in the next section such as `yyyyMMddHHmmss`
File format	Text field delimiter	This optional parameter is used when a delimiter is used for text fields. This is especially useful when the CSV contains text data with carriage returns.
File format	Minimum nb of records required	This parameter indicates the minimum number of records that must be present in the CSV file to consider this file as valid. Thanks to this parameter you can detect errors such as empty dumps

Here is the description of the date format syntax:

Letter	Date or Time Component	Presentation	Examples
G	Era designator	Text	AD
y	Year	Year	1996; 96
Y	Week year	Year	2009; 09
M	Month in year	Month	July; Jul; 07
w	Week in year	Number	27
W	Week in month	Number	2
D	Day in year	Number	189
d	Day in month	Number	10
F	Day of week in month	Number	2

Letter	Date or Time Component	Presentation	Examples
E	Day name in week	Text	Tuesday; Tue
u	Day number of week (1 = Monday, . . . , 7 = Sunday)	Number	1
a	Am/pm marker	Text	PM
H	Hour in day (0-23)	Number	0
k	Hour in day (1-24)	Number	24
K	Hour in am/pm (0-11)	Number	0
h	Hour in am/pm (1-12)	Number	12
m	Minute in hour	Number	30
s	Second in minute	Number	55
S	Millisecond	Number	978
z	Time zone	General time zone	Pacific Standard Time; PST; GMT-08:0
Z	Time zone	RFC 822 time zone	-0800
X	Time zone	ISO 8601 time zone	-08; -0800; -08:00

Here are some examples:

Date and Time Pattern	Result
“yyyy.MM.dd G ‘at’ HH:mm:ss z”	2001.07.04 AD at 12:08:56 PDT
“EEE, MMM d, ’ ’yy”	Wed, Jul 4, ’01
“h:mm a”	12:08 PM
“hh ‘o’‘clock’ a, zzzz”	12 o’clock PM, Pacific Daylight Time
“K:mm a, z”	0:08 PM, PDT
“yyyyy.MMMMM.dd GGG hh:mm aaa”	02001.July.04 AD 12:08 PM
“EEE, d MMM yyyy HH:mm:ss Z”	Wed, 4 Jul 2001 12:08:56 -0700
“yyMMddHHmmssZ”	010704120856-0700
“yyyy-MM-dd’T’HH:mm:ss.SSSZ”	2001-07-04T12:08:56.235-0700
“yyyy-MM-dd’T’HH:mm:ss.SSSXXX”	2001-07-04T12:08:56.235-07:00
“YYYY-’W’ww-u”	2001-W27-3

File columns The validation follows a strict strategy regarding the columns of the file:

- All columns presents in the file must be declared.
- Columns must be named (first line of the file)
- Columns order can’t be changed

If one of those prerequisites is not met, the file is not considered valid.

When you declare a new column, the following fields are available:

Name	Description
Name	The name of the column as it appears in the file
Type	The type of value in the column, can be String, Date, Custom date, Boolean, Number, Email. If the type is set to Date or Boolean, the configuration in the previous section is used to check the values. If the type is set to custom date, a special date format must be provided in the next parameter (format). If the type is email, the email address validity is checked against RFC 5322. The Number type expects Integer values. If you consider this as too strict, you should rather select String a your column type and write your own regex in the next field.
Format	When the type is set to String, you can declare a regex here to check the validity of the values. When the type is set to custom date, you can declare a date format here to check the validity of the values.
Format description	This parameter is used with the previous one when the column type is set to String. In case of an error, instead of displaying the expected regex, we use this value to inform the application owner that an error has been found in its file.
Content must be unique	Check this box if the column value must be unique in the file
Content is mandatory	Check this box if a content is mandatory in this column
Minimum fill factor	When the content is not mandatory, you can declare here a minimum fill factor, an error will be triggered if the minimum fill factor is not reached at the end of the file. For instance if you enter 30, it means that at least 30% of the rows must contain a value in this column. This is useful when a content is not mandatory BUT expected in order to detect empty columns.
Order	The column order in the CSV file, starting by 1

Add computed columns

If needed you can add dynamic (computed) columns on the fly on your datasource.

This can be useful when you want to:

- add static information in your mapping
- compute dynamic information based on the current CSV line

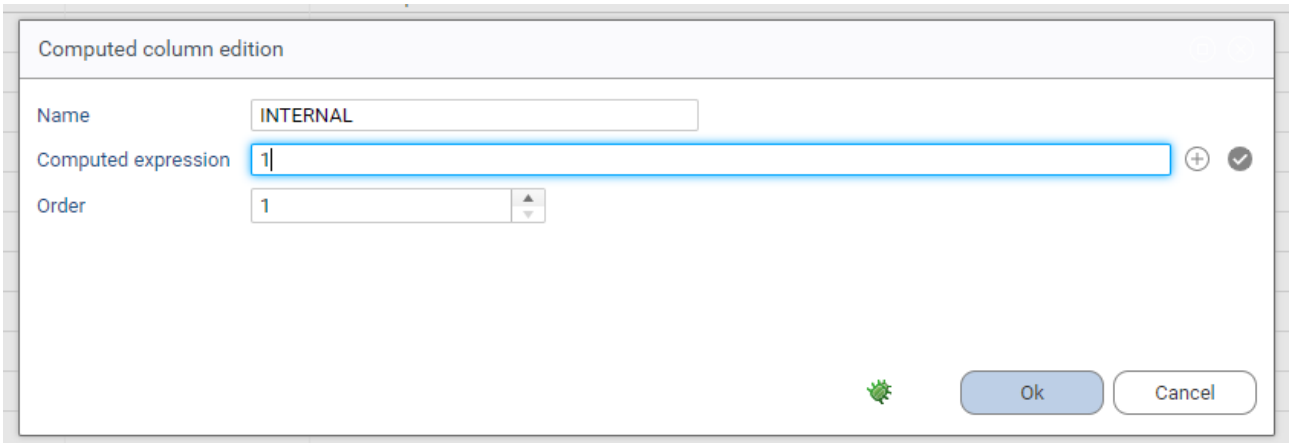
Let’s take some examples:

- You have a CSV HR file for internal people, therefore you **know** that 100% of the identities located in this file are internal people, nevertheless this information is not located in the file itself. You can add a computed column in the form of `INTERNAL=True` in order to be able to map this information in the Brainwave datamodel.

- You have a CSV HR file with `firstname` and `lastname` and you want to map the `fullname` column of the Brainwave datamodel, therefore, you want to add a computed column in the form of `FULLNAME=CSVFIRSTNAME+''+CSVLASTNAME` where `CSVFIRSTNAME` and `CSVLASTNAME` are the current values in the CSV file.
- You have a CSV HR file and you know that depending on the `employee number` syntax you can spot if the identity is internal or a contractor (let's say for instance that he is an internal people if the employee number starts with `I`). You will want to create a computed column in the form of `If employeenumber starts with 'I'Then INTERNAL=True Else INTERNAL=False`.

All those use cases are covered thanks to the `Add computed columns` section.

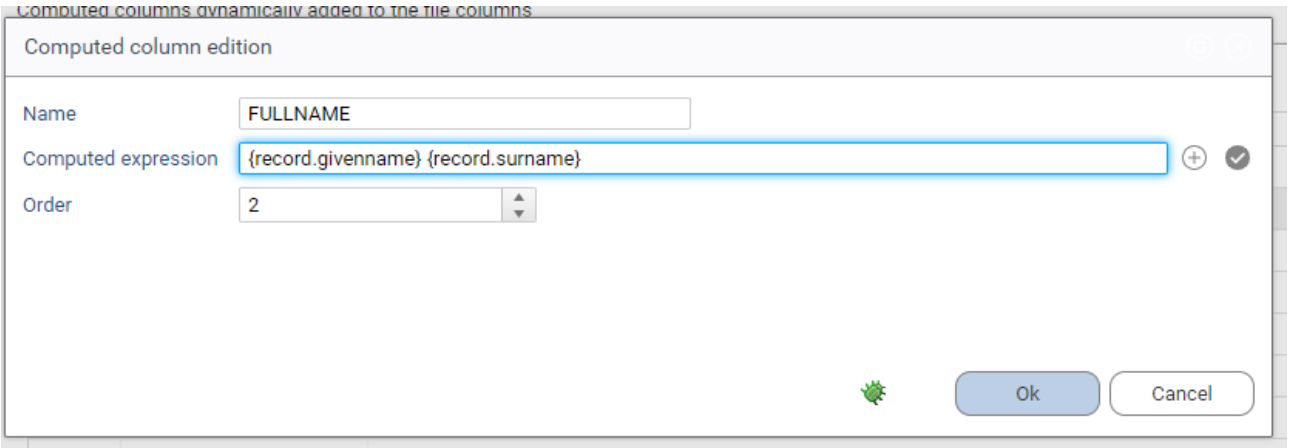
To add a computed column, click on `Add a new computed column` and provide a name and a value in the form of a computed expression:



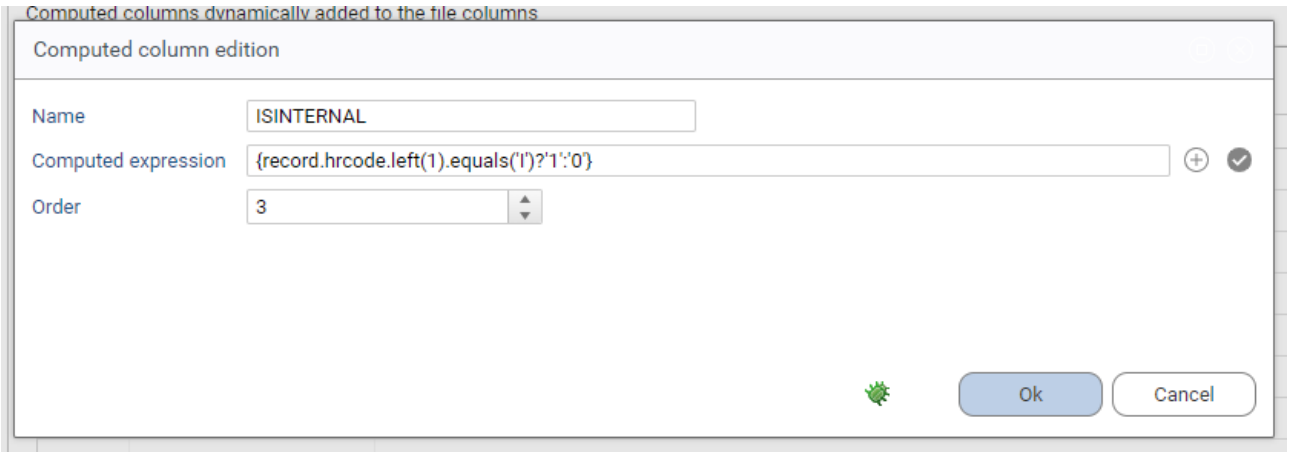
A computed expression is computed for each line of your CSV file. It is a simple String with escape sequences whenever you want to refer to other columns of the current line.

> **Important:** A computed expression **always** returns a String. If you want to return a Boolean value with a computed expression, you must return string character `0` for False and string character `1` for True. If you want to return a Date value with a computed expression, you must return it as a String in LDAP format `yyyyMMddHHmmss`.

The easiest way to add a reference to another column of the file is to click on the + icon next to the computed expression field.
You can mix both static content and dynamic content in your computed expression.



As you can see, a reference to another column is in the form of `{record.COLUMNNAME}`:



The expression within `{ }` follows a server side javascript syntax. record.COLUMNNAME is provided as String. You can add transform function to your record. If you want to return a value depending on a condition the best practice is to use ternary operator `CONDITION?VALUEIFTRUE:VALUEIFFALSE`.

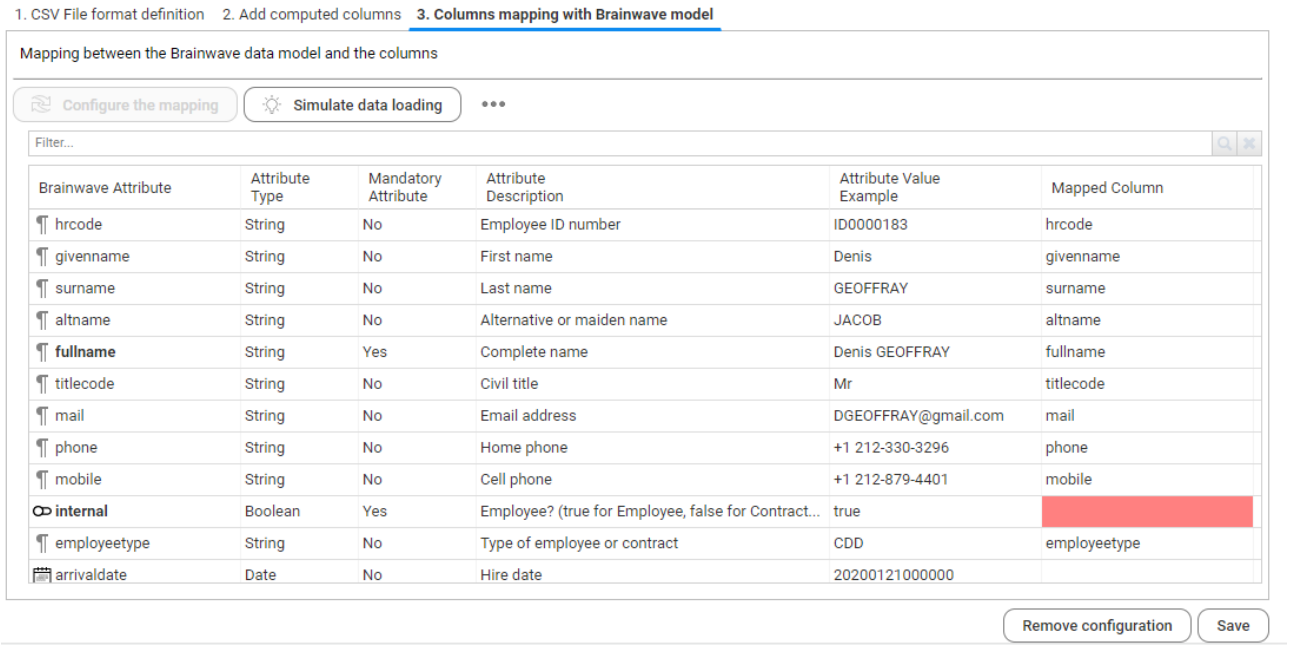
> **Important:** A record is always provided in a form of a String even if the CSV column definition is a number, date, boolean, … In case of a boolean you will have `0` if False and `1` if true. In cas of a date you will have an LDAP string representation of the date `yyyyMMddHHmmss`.

> **Important:** Computed column is an advanced feature. Don't forget to validate your javascript expression as your configuration won't work in case of error.

Columns mapping with Brainwave model

Once you have declared your CSV columns and your computed columns the only thing you have to do is to map those columns to the Brainwave data model. This is done through the last tab.

All expected columns are listed in the table along with the their type and an example. In order to map a column you just have to click on the corresponding `Brainwave Attribute` in the table.
Once the mapping is done, the CSV column name will appear in `Mapped Column`:

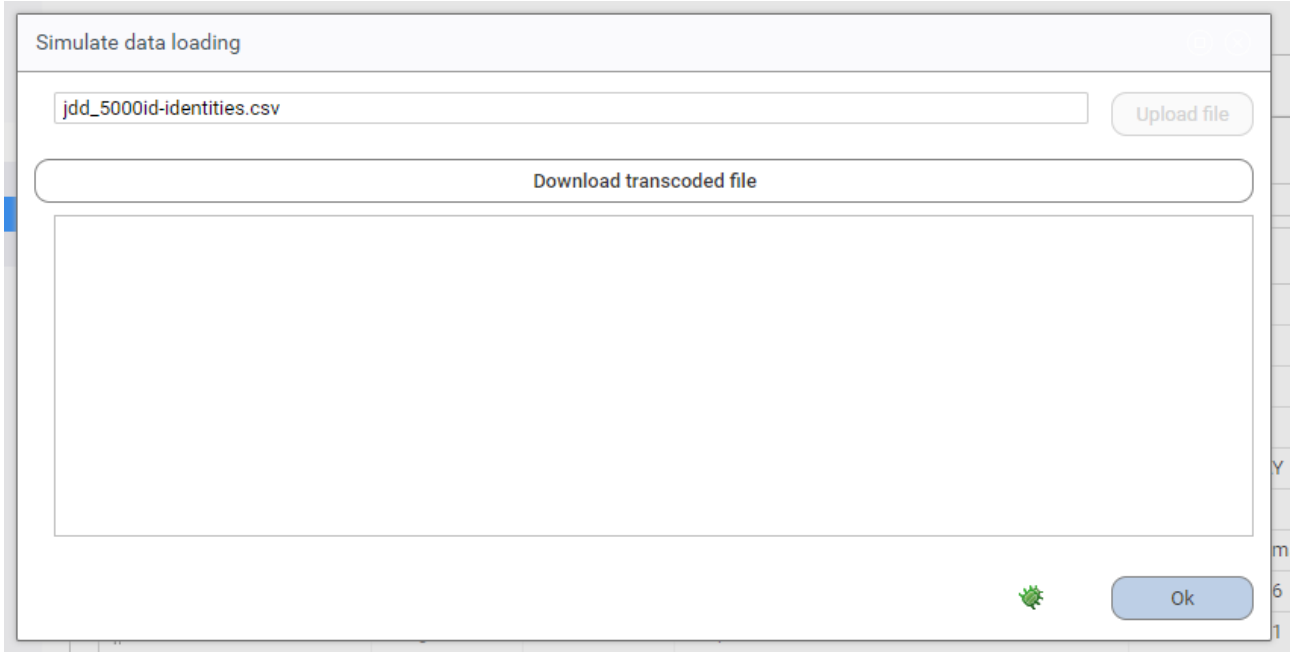


Brainwave Attribute	Attribute Type	Mandatory Attribute	Attribute Description	Attribute Value Example	Mapped Column
hrcode	String	No	Employee ID number	ID0000183	hrcode
givenname	String	No	First name	Denis	givenname
surname	String	No	Last name	GEOFFRAY	surname
altname	String	No	Alternative or maiden name	JACOB	altname
fullname	String	Yes	Complete name	Denis GEOFFRAY	fullname
titlecode	String	No	Civil title	Mr	titlecode
mail	String	No	Email address	DGEOFFRAY@gmail.com	mail
phone	String	No	Home phone	+1 212-330-3296	phone
mobile	String	No	Cell phone	+1 212-879-4401	mobile
internal	Boolean	Yes	Employee? (true for Employee, false for Contract...	true	
employeetype	String	No	Type of employee or contract	CDD	employeetype
arrivaldate	Date	No	Hire date	20200121000000	

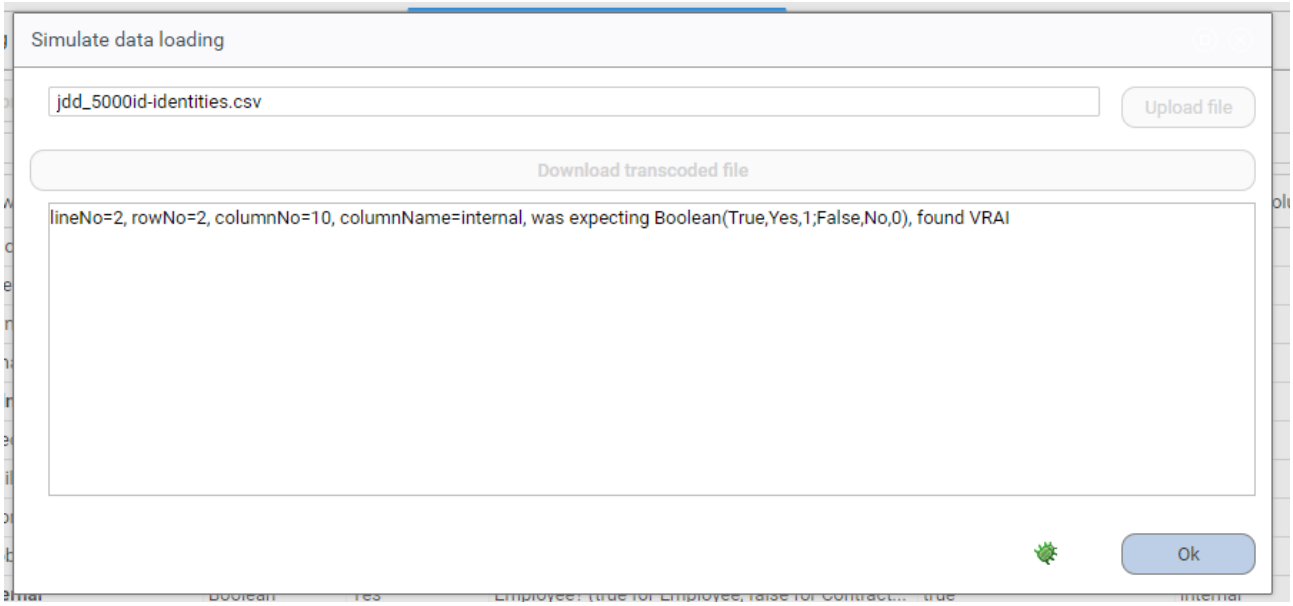
> **Important:** `Mapped Columns` background field is red In case of mandatory columns.

> **Note** If your input file complies with Brainwave pivot format, the mapping will be performed automatically. In order for this to happen both columns names and columns types must be the same.

Once your mapping is done you can test it. In order to do it click on `Simulate data loading` and pick your CSV file.
The file will be uploaded, parsed and transcoded to a `pivot` format. You are **strongly encouraged** to download the transcoded file in order to check if both your mappings and your computed values are correct.

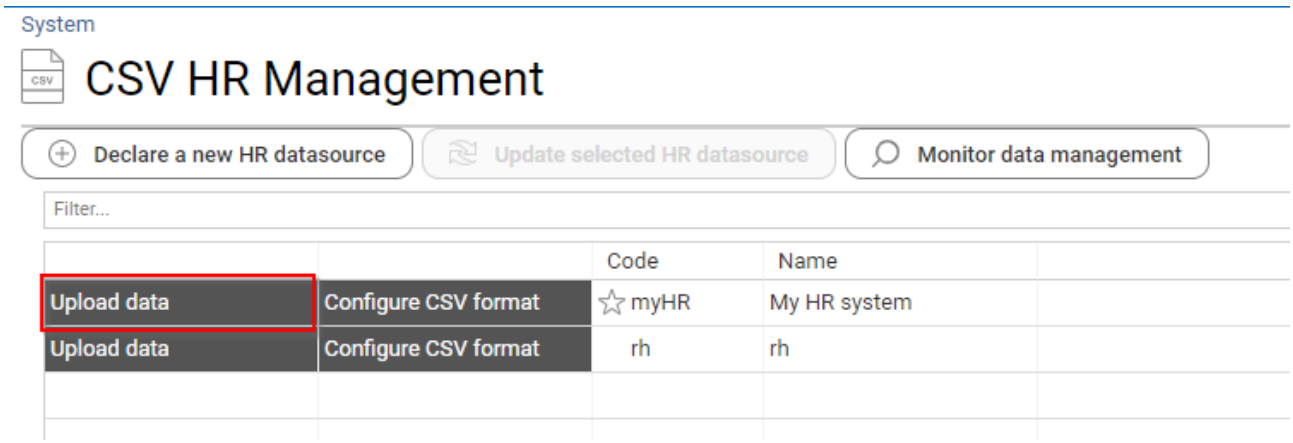


In case of configuration error, a message will appear in the dialog box:



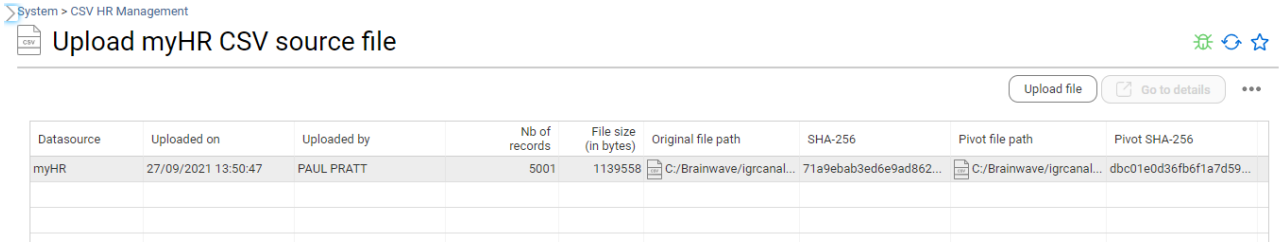
Upload data

Once your datasource is properly configured, you can upload data as a technical administrator. In order to do so, go to `CSV HR`/`Application Management` and click `Upload data` on the corresponding datasource.

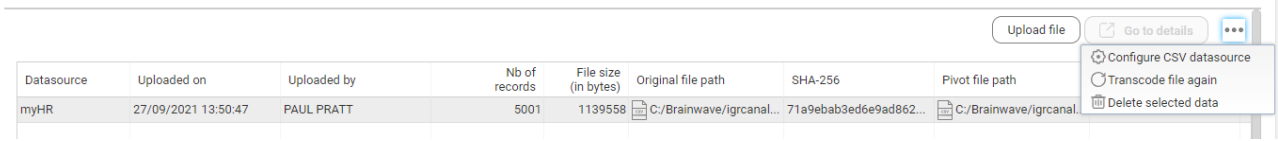


You can then upload fresh data thanks to the `Upload file` button.

Once your data is uploaded it with appear in the list. You can come back to the page at a later time if you want to download the original file or the transformed (pivot) file.



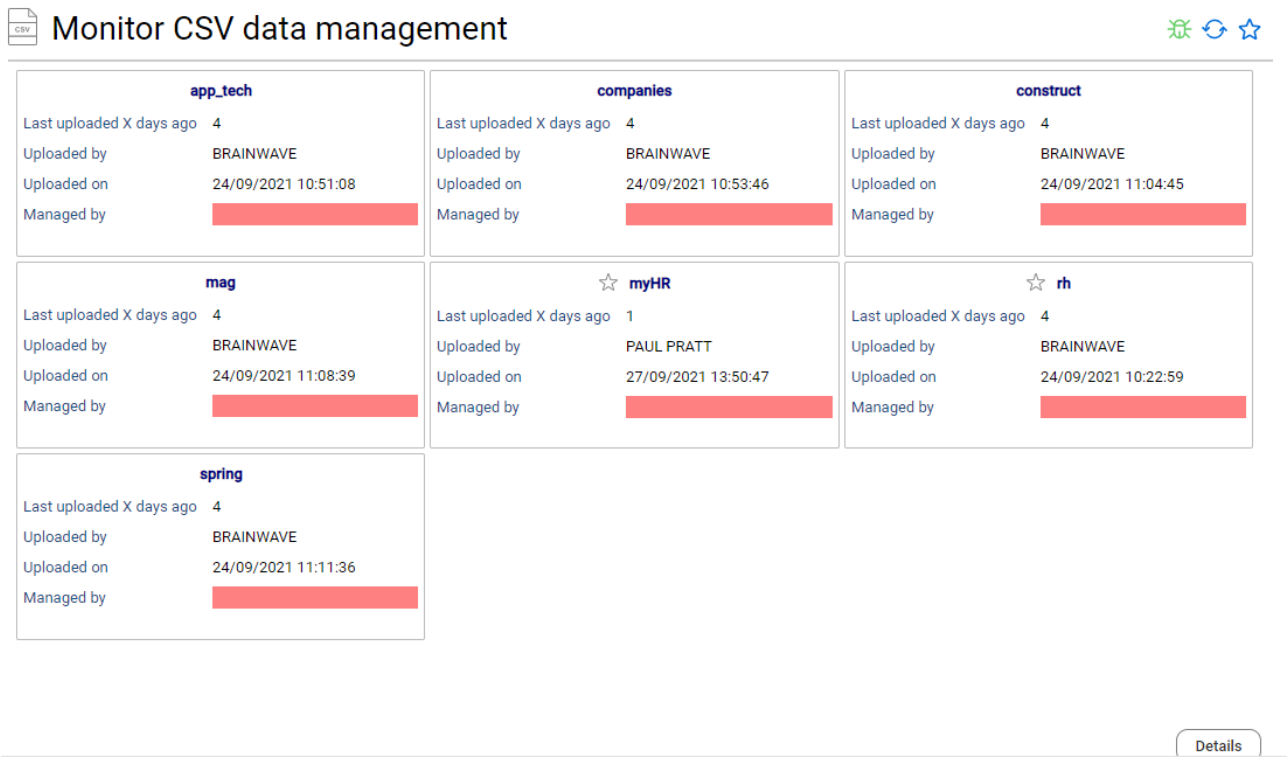
If you have updated your configuration, you can transcode the file again. In order to do this, select the line and click on `Transcode file again` in the contextual menu.



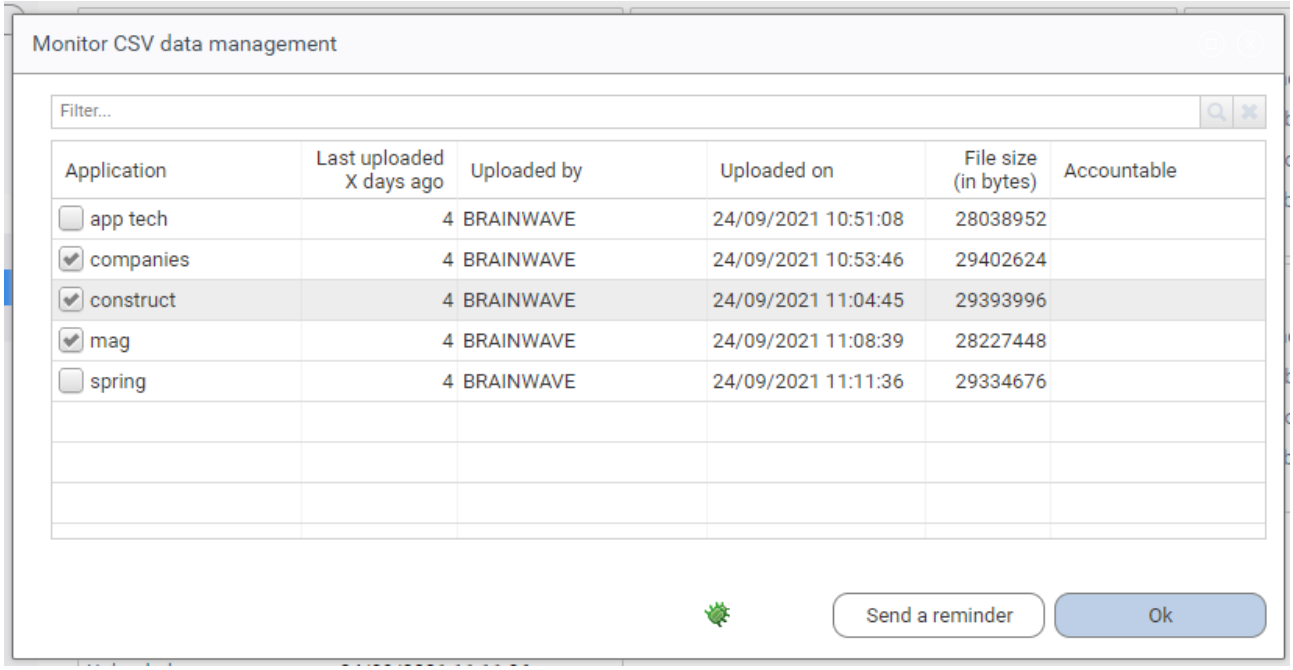
> **Important:** The next time you will refresh the Brainwave data model, the most recent data will be used for each datasource.

Monitor data management

You can have an overview of the current status for all your datasources through the `Monitor data management` button located in the `CSV HR`/`Application Management`:



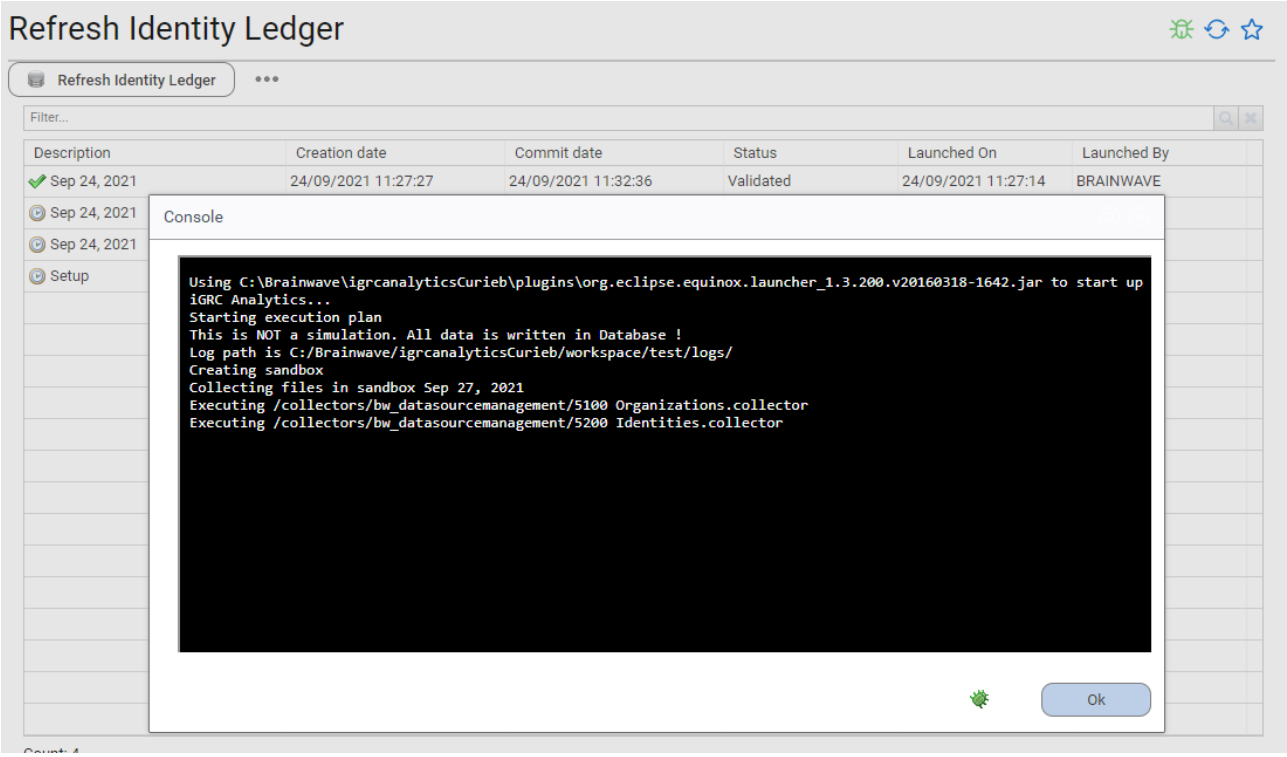
You can click on `Details` for an alternate view and if you want to send gentle reminders by email to the application owners.



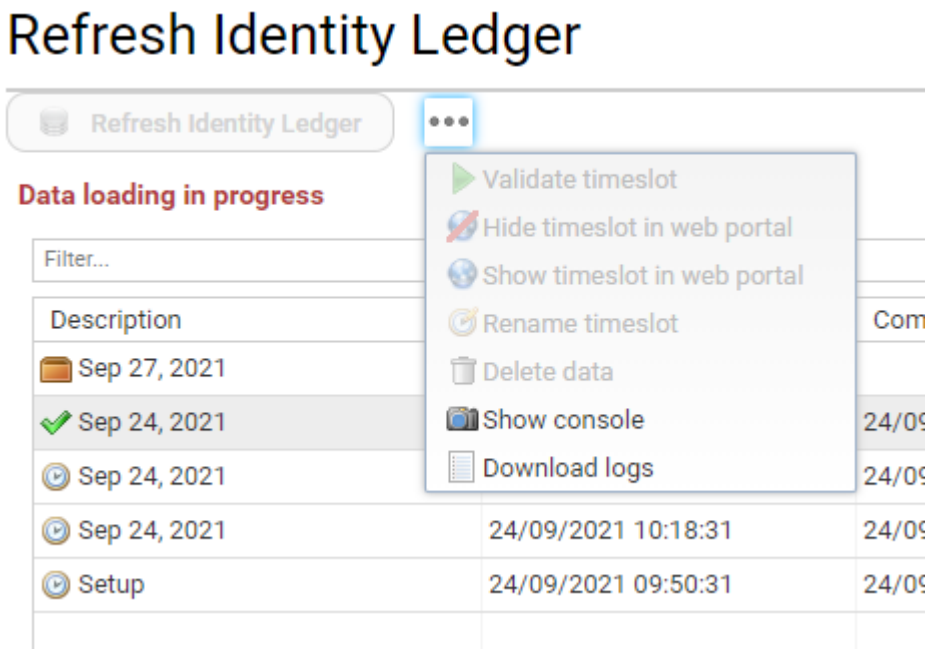
Refresh data

This add-on comes with a special feature in order to allow you to refresh your Identity Ledger from the web interface.

In order to refresh your Identity Ledger, go to `Settings`/`System` and select `Refresh Identity Ledger` :



Through this interface you can also download logs in case of errors:



> **Important:** Once refreshed, you **must** logoff and logon to connect to the latest timeslot.

Prerequisites

In order for this to work Brainwave must be installed and configured on a **single server**.

- Both Apache Tomcat and batch data loading **must** run on the same server
- No failover / cluster support

You **must** configure the following technical variables:

Variable name	Variable description	Typical value
datasourcemanagement_loadpath	iGRC path to load data	C:/Program Files/igrcanalytics/
datasourcemanagement_loadbatch	batch to load data	igrc_batch.cmd
datasourcemanagement_loadprojectname	project name to load data	
datasourcemanagement_loadtechnical	technical configuration to use to load data	
datasourcemanagement_loadconfigpath	configuration path to use to load data	C:/Program Files/igrcanalytics/

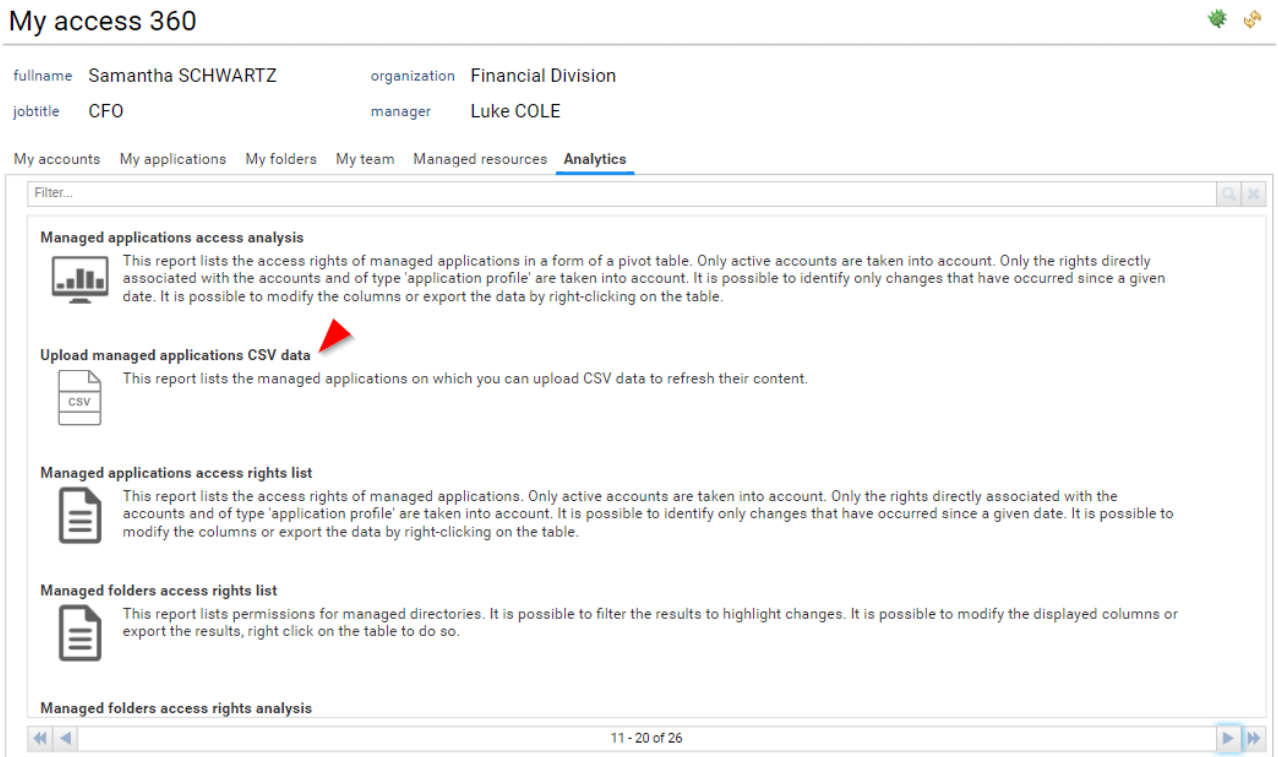
Self-service data upload

Once the configuration is set up, you can use the add-on to upload your application CSV file. The add-on is available as a dedicated analytics of the application. Therefore, the upload page is available to:

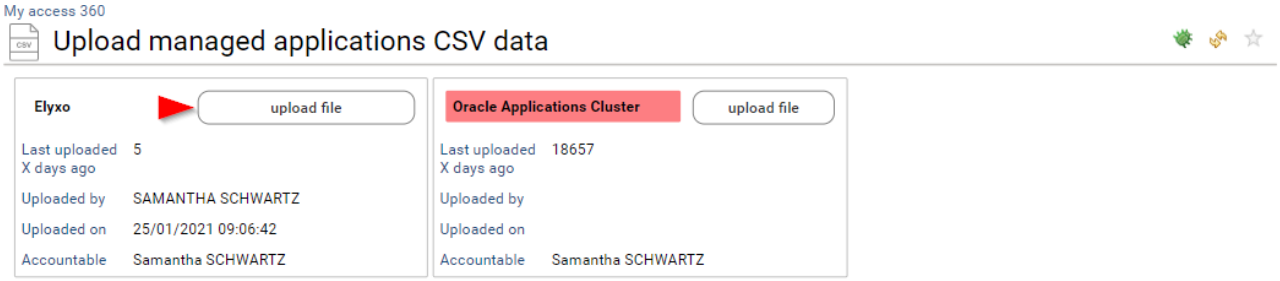
- technical administrators
- functional administrators
- application owners (both business & technical owners)

Let's connect as an application business owner and see what's happens.

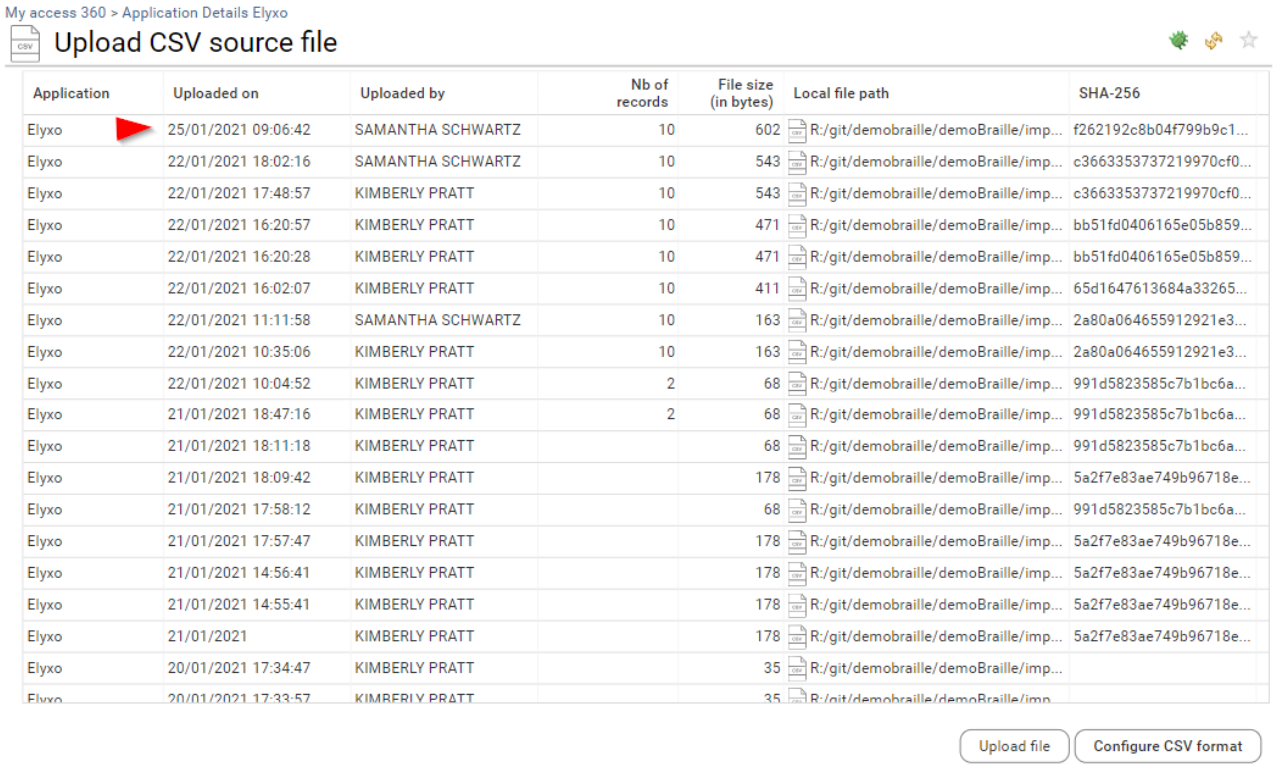
Once connected, the application business owner can access to the CSV management page for its applications through the `Analytics` pane.



This page presents all the application on which the current user can upload fresh data. The user just needs to click on `upload file`.

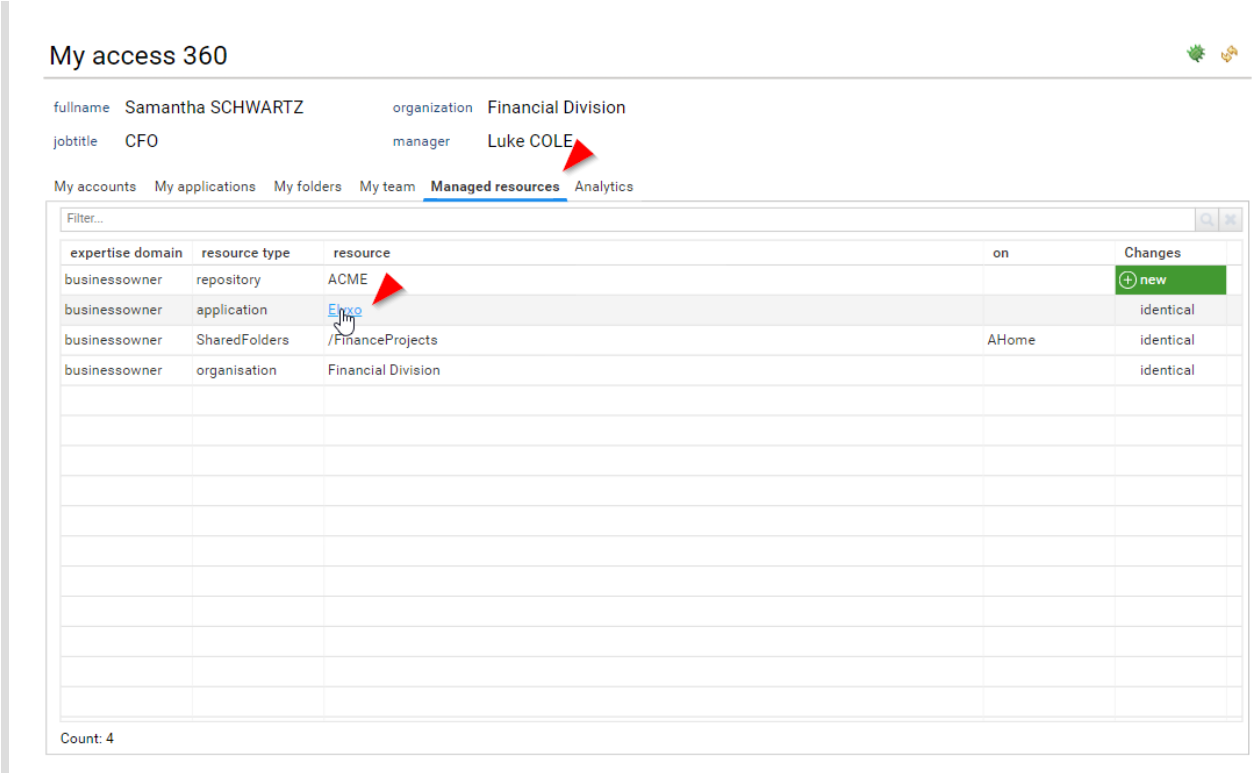


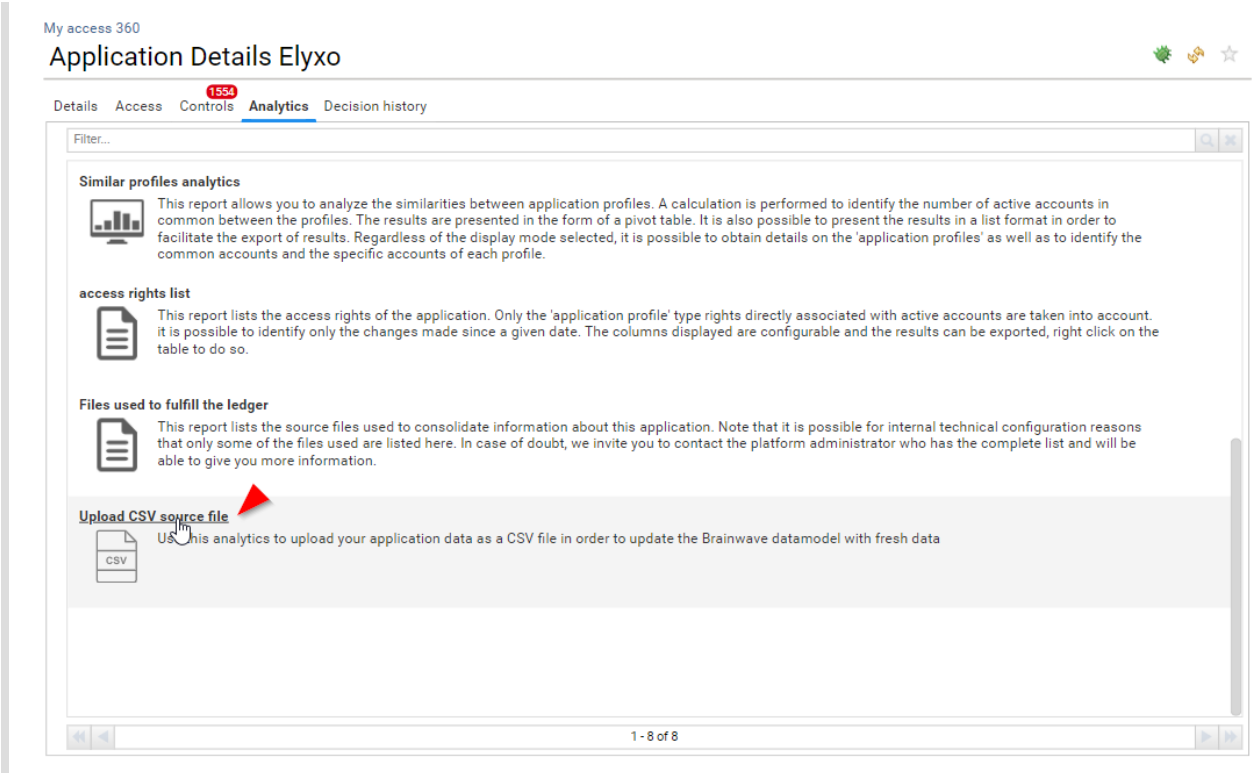
He is then redirect to the selected application CSV upload page. He just have to click on `Upload file` to upload the file and to see it appearing in the list.



Application	Uploaded on	Uploaded by	Nb of records	File size (in bytes)	Local file path	SHA-256
Elyxo	25/01/2021 09:06:42	SAMANTHA SCHWARTZ	10	602	R:/git/demobraille/demoBraille/imp...	f262192c8b04f799b9c1...
Elyxo	22/01/2021 18:02:16	SAMANTHA SCHWARTZ	10	543	R:/git/demobraille/demoBraille/imp...	c3663353737219970cf0...
Elyxo	22/01/2021 17:48:57	KIMBERLY PRATT	10	543	R:/git/demobraille/demoBraille/imp...	c3663353737219970cf0...
Elyxo	22/01/2021 16:20:57	KIMBERLY PRATT	10	471	R:/git/demobraille/demoBraille/imp...	bb51fd0406165e05b859...
Elyxo	22/01/2021 16:20:28	KIMBERLY PRATT	10	471	R:/git/demobraille/demoBraille/imp...	bb51fd0406165e05b859...
Elyxo	22/01/2021 16:02:07	KIMBERLY PRATT	10	411	R:/git/demobraille/demoBraille/imp...	65d1647613684a33265...
Elyxo	22/01/2021 11:11:58	SAMANTHA SCHWARTZ	10	163	R:/git/demobraille/demoBraille/imp...	2a80a064655912921e3...
Elyxo	22/01/2021 10:35:06	KIMBERLY PRATT	10	163	R:/git/demobraille/demoBraille/imp...	2a80a064655912921e3...
Elyxo	22/01/2021 10:04:52	KIMBERLY PRATT	2	68	R:/git/demobraille/demoBraille/imp...	991d5823585c7b1bc6a...
Elyxo	21/01/2021 18:47:16	KIMBERLY PRATT	2	68	R:/git/demobraille/demoBraille/imp...	991d5823585c7b1bc6a...
Elyxo	21/01/2021 18:11:18	KIMBERLY PRATT		68	R:/git/demobraille/demoBraille/imp...	991d5823585c7b1bc6a...
Elyxo	21/01/2021 18:09:42	KIMBERLY PRATT		178	R:/git/demobraille/demoBraille/imp...	5a2f7e83ae749b96718e...
Elyxo	21/01/2021 17:58:12	KIMBERLY PRATT		68	R:/git/demobraille/demoBraille/imp...	991d5823585c7b1bc6a...
Elyxo	21/01/2021 17:57:47	KIMBERLY PRATT		178	R:/git/demobraille/demoBraille/imp...	5a2f7e83ae749b96718e...
Elyxo	21/01/2021 14:56:41	KIMBERLY PRATT		178	R:/git/demobraille/demoBraille/imp...	5a2f7e83ae749b96718e...
Elyxo	21/01/2021 14:55:41	KIMBERLY PRATT		178	R:/git/demobraille/demoBraille/imp...	5a2f7e83ae749b96718e...
Elyxo	21/01/2021	KIMBERLY PRATT		178	R:/git/demobraille/demoBraille/imp...	5a2f7e83ae749b96718e...
Elyxo	20/01/2021 17:34:47	KIMBERLY PRATT		35	R:/git/demobraille/demoBraille/imp...	
Elyxo	20/01/2021 17:33:57	KIMBERLY PRATT		35	R:/git/demobraille/demoBraille/imp...	

> **Note:** The user can also access to the upload page through the `Managed resources` pane by selecting the application and then selecting `Upload CSV source file` in the `Analytics` pane.





Once the file is uploaded, it automatically appears in the list.

The following metadata are associated with the file and are presented in this table:

Field	Description
Application	The application name
Uploaded on	Date&time when the file has been uploaded

Field	Description
Uploaded by	Fullname of the individual who has uploaded the file
Nb of records	Number of records found in the file
File size	File size (in bytes)
Local file path	Location of the uploaded file in the webserver
SHA256	Sha256 Hash of the uploaded file

Thanks to those information, you can maintain a chain of custody related to the data used to fulfill the Brainwave identity ledger.

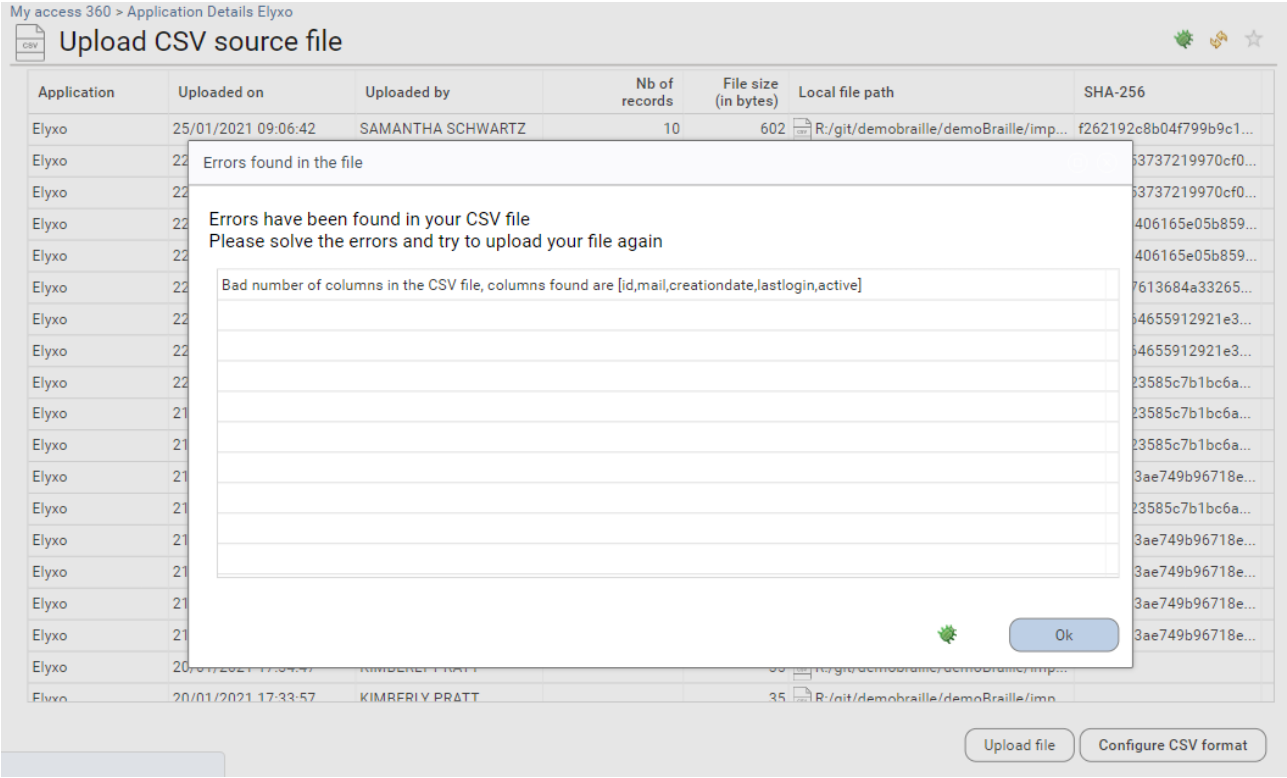
> **Tips:** If you want to double-check that the file has not been modified since it has been uploaded by the application owner, you can use the SHA256. On windows system, the following command line will help you to compute a SHA256 has of your file `certutil -hashfile myfile.csv SHA256` Use this command line to compute the SHA256 hash and compare it with the one stored in Brainwave.

As you can see, the file is stored in the location specified through the `datasourcemanagement_location` configuration variable.
The naming convention of the file is:

```
APPLICATIONCODE_yyyMMMddHHmmss.csv
```

Where `APPLICATIONCODE` is the application unique code and `yyyyMMMddHHmmss` represents the date&time when the file has been uploaded.

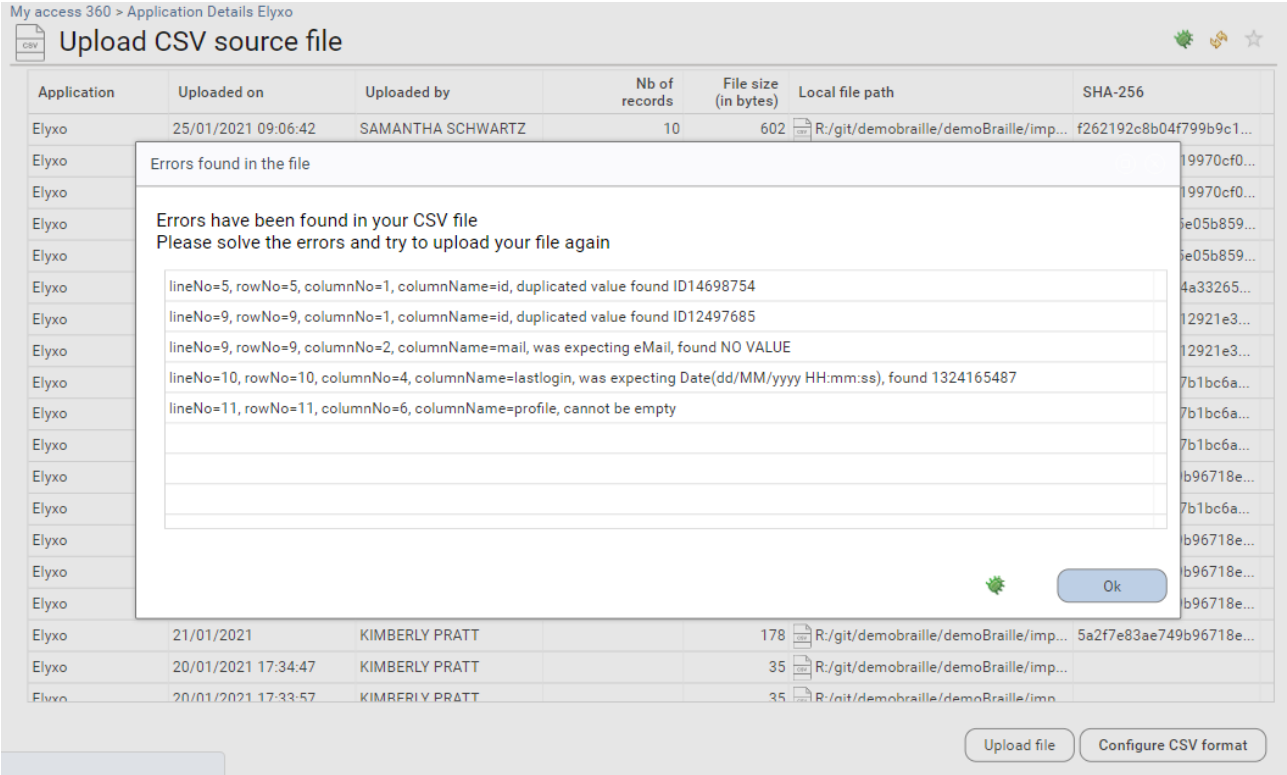
What happens when the file is incomplete/corrupted/non compliant? Let's try to upload a file with some missing columns and see what's happens:



As you can see, although the file is uploaded to a temporary folder and checked, as some errors are found a status is displayed to the application owner and the file is not copied in `datasourcemanagement_location`.

> **Warning:** Even if the user interfaces can be nationalized, the errors will ALWAYS be displayed in plain english.

Here is another example:



As you can see, when errors are detected in the content, they are listed in the table.

`lineNo` represents the line number in the file while `rowNo` represents the record number in the file. They can be different if some text field values contain carriage returns.

If more than 100 errors are found in the file, the check is aborted with this final error message `Too many errors, abort file format analysis`.

Technical corner

For advanced configuration, you will find here a series of information in order to help you to fine tune the add-on.

Retrieve information through your own Views

All information related to CSV configuration and uploaded files are stored as a series of Metadata. Those metadata are located in `{project path}\metadata\bw_datasourcemanagement\`:

- **datasource.metadata** is associated with uploaded data

- **datasourceformat.metadata** is associated with the CSV file general configuration
- **datasourcecolumn.metadata** is associated with each declared column in the configuration

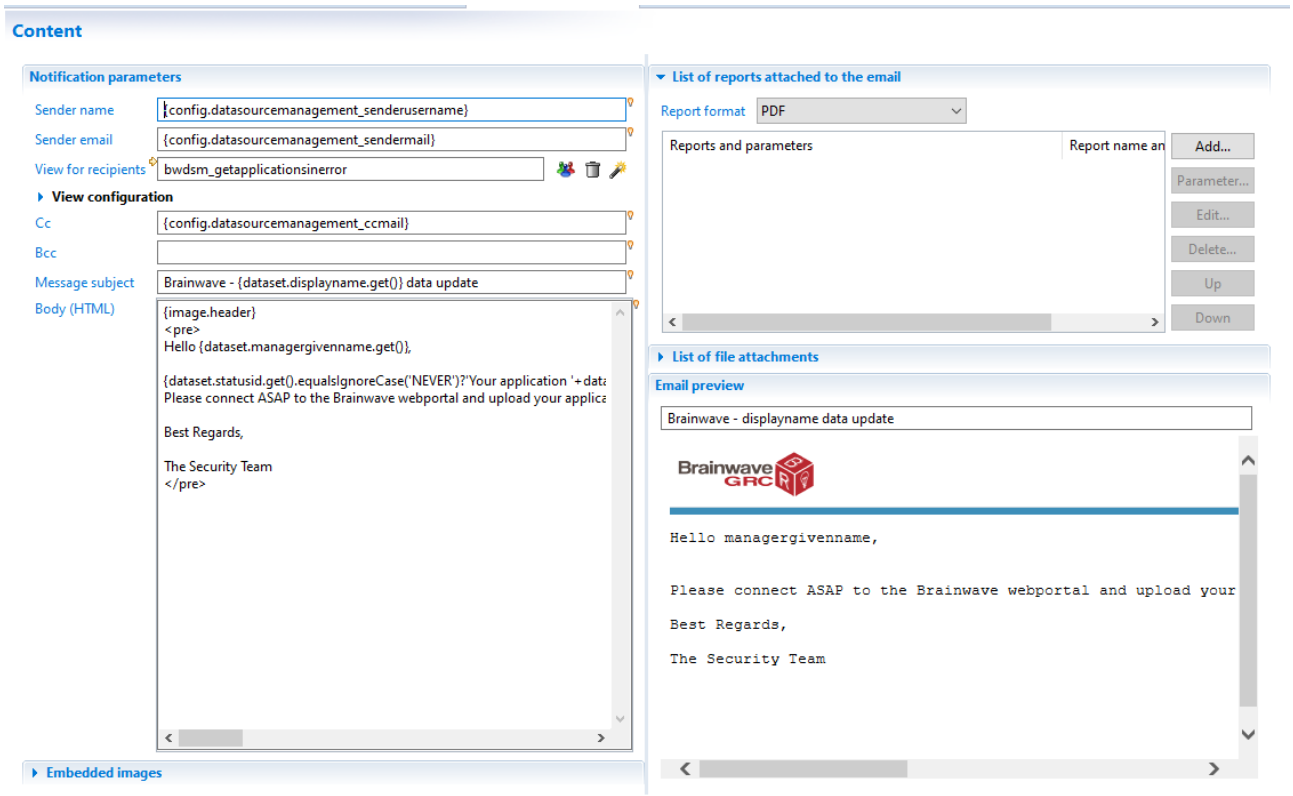
All the information are attached to the application as a set of attributes. Here is a summary of those attributes:

Attribute	Description
`bw_datasource_location`	Absolute location of the uploaded file
`bw_datasource_by`	Fullname of the individual who has uploaded the file
`bw_datasource_size`	Size of the uploaded file in bytes
`bw_datasource_nbrows`	Number of records in the uploaded file
`bw_datasource_upload_date`	Date and time when the file was uploaded
`bw_datasource_hash`	SHA256 hash of the file
`bw_datasourceformat_separator`	Character used as a separator in the CSV file
`bw_datasourceformat_delimiter`	String delimiter used in the CSV file
`bw_datasourceformat_minimumnblines`	Minimum number of lines to consider the file valid
`bw_datasourceformat_errorifolderused`	used by gentle reminder, error threshold
`bw_datasourceformat_warningifolderused`	used by gentle reminder, warning threshold
`bw_datasourceformat_checkfilesizevariation`	Activate file size check
`bw_datasourceformat_filesizevariation`	Allowed file size variation when file size check is activated
`bw_datasourceformat_encoding`	CSV file encoding
`bw_datasourceformat_booleanformat`	Default boolean format in the file
`bw_datasourceformat_dateformat`	Default date format in the file
`bw_datasourcecolumn_name`	CSV column name
`bw_datasourcecolumn_type`	CSV column type
`bw_datasourcecolumn_order`	CSV column order
`bw_datasourcecolumn_minimumfillfactor`	CSV column minimum fill factor
`bw_datasourcecolumn_mandatory`	CSV column is mandatory
`bw_datasourcecolumn_format`	CSV column format
`bw_datasourcecolumn_formatprettystring`	CSV column format description

Attribute	Description
`bw_datasourcecolumn_mustbeunique`	CSV column content must be unique

Fine tune the gentle reminder email content

Gentle reminders are configured through a notification rule located in `{project path}/notifications/bw_datasourcemanagement/reminder.notifyrule`.



If you update it, don't forget to update the nationalized versions accordingly.

Automate the sending of gentle reminders

The `{project path}/notifications/bw_datasourcemanagement/reminder.notifyrule` will only send emails when relevant:

- A file has never been provided
- The file is older than the configuration warning threshold
- The file is older than the configuration error threshold

As a result, you can execute this notification rule on a regular basis without any side effect.

In order to do so, you just have to cron/batch the sending of the email from your server.

Please consult Sending emails in batch mode for more information about sending notification emails through the command line.

Customize the uploaded file name and file location

In order to customize the file naming convention as well as its location you can update the following workflow `{project path}/workflow/bw_datasourcemanagement/uploaded.workflow`.
This can be useful for instance if you want to directly replace the current application file in a way that the next execution plan launch will leverage it.

> **Warning:** This is an advanced customization, you can perform it but at your own risks.

Fine tune access rights

Access rights to access the datasource managements component are configured under `/webportal/features/bw_datasourcemanagement/`.

The following features are available:

Feature name	Feature description
`bw_datasourcemanagement_canupdateconfig`	can update configuration
`bw_datasourcemanagement_cansendreminder`	can send a gentle reminder through the monitoring page
`bw_datasourcemanagement_canuploadfile`	can upload CSV file
`bw_datasourcemanagement_uploadanalytics`	Shows the upload analytics page

The following featureset are available and configured like this by default:

Featureset name	Featureset description	User	Campaign Manager	Function Admin	Admin	Technical admin
`bw_datasourcemanagement_canmanageconfiguration`	can manage configuration					X
`bw_datasourcemanagement_canaccess`	can access the upload analytics page	X	X	X	X	X

By default, `bw_datasourcemanagement_uploadanalytics` is available for both featuresets while the rest is available for `bw_datasourcemanagement_canmanageconfiguration` only.